DSTL (BCS-303)
ST-2 Solutions

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Q1(a)

i) Find the converse, inverse and contrapositive of the assertion
"I stay only if you go". [GATE CSE 2001]

ii) Convert the assertion below into symbolic form using predicates and quantifiers:
"Not all rainy days are cold". [GATE CSE 2014]

Solution

i) Converse / inverse / contrapositive
Let

$$p : \text{"I stay"}, \qquad q : \text{"you go"}.$$

The statement "I stay only if you go" means

$$p \to q.$$

- **Converse**: $q \to p$ ("If you go, then I stay").
- **Inverse**: $\neg p \to \neg q$ ("If I do not stay, then you do not go").
- **Contrapositive**: $\neg q \to \neg p$ ("If you do not go, then I do not stay").

ii) Predicate–logic form
Domain: all days.
Predicates:

$$R(x) : \text{"}x \text{ is rainy"}, \qquad C(x) : \text{"}x \text{ is cold"}.$$

"Not all rainy days are cold" means:

$$\neg\forall x \left(R(x) \to C(x)\right)$$

Equivalently (pushing the negation in):

$$\exists x \left(R(x) \land \neg C(x)\right).$$

So a correct symbolic form is

$$\boxed{\exists x \left(R(x) \land \neg C(x)\right)}.$$

Q1(b)

Let p and q be two propositions. Consider the following formulae in propositional logic:

$$\text{S1} : (\neg p \land (p \lor q)) \to q, \qquad \text{S2} : q \to (\neg p \land (p \lor q)).$$

Which of the following correctly identifies each statement as a *tautology*, *contradiction*, or *contingency*? [GATE CSE 2021]

Solution

We simplify each formula.

For S1:

$$\neg p \land (p \lor q) \equiv (\neg p \land p) \;\lor\; (\neg p \land q) \equiv \mathbf{F} \lor (\neg p \land q) \equiv \neg p \land q.$$

Thus

$$S1 \equiv (\neg p \wedge q) \to q.$$

If $(\neg p \wedge q)$ is true, then q is of course true, so the implication is true. If $(\neg p \wedge q)$ is false, the implication is true by definition. Hence S1 is *always* true.

$$\boxed{\text{S1 is a tautology.}}$$

For S2:

$$\neg p \wedge (p \vee q) \equiv \neg p \wedge q$$

(as above). Then

$$S2 \equiv q \to (\neg p \wedge q) \equiv \neg q \vee (\neg p \wedge q).$$

Use distributivity:

$$\neg q \vee (\neg p \wedge q) \equiv (\neg q \vee \neg p) \wedge (\neg q \vee q) \equiv (\neg q \vee \neg p) \wedge \mathbf{T} \equiv \neg q \vee \neg p \equiv \neg(p \wedge q).$$

The formula $\neg(p \wedge q)$ is true for some valuations and false for others, so it is neither always true nor always false.

$$\boxed{\text{S2 is a contingency.}}$$

Q2(a)

Show that t is a valid conclusion from the given premises:

$$\neg p \wedge q, \quad r \to p, \quad \neg t \to s, \quad s \to t.$$

Solution

We must derive t using rules of inference.

1. $\neg p \wedge q$ (Premise)
2. $\neg p$ (From 1 by *simplification*)
3. $r \to p$ (Premise)
4. $\neg r$ (From 2 and 3 by *modus tollens*)

The variable r is actually irrelevant for deriving t; the key is the pair of premises involving t:

5. $\neg t \to s$ (Premise)
6. $s \to t$ (Premise)

Now argue by *indirect proof*:

7. Assume $\neg t$. (Assumption for contradiction)
8. From 5 and 7: s. (Modus ponens)
9. From 6 and 8: t. (Modus ponens)
10. From 7 and 9 we obtain a contradiction: $t \wedge \neg t$.

Therefore our assumption $\neg t$ is impossible, so t must be true:

$$\boxed{t}$$

Hence t is a valid conclusion from the premises.

Q2(b)

Determine whether the given proposition is a tautology, contradiction or contingency:

$$\big(q \vee \neg(p \wedge r)\big) \rightarrow \big((q \vee \neg p) \vee \neg r\big).$$

Solution

Let

$$A = q \vee \neg(p \wedge r), \qquad B = (q \vee \neg p) \vee \neg r.$$

We must check whether $A \rightarrow B$ can ever be false.
An implication is false only when its antecedent is true and its consequent is false.

Step 1: Force the consequent to be false.
B is false iff each of its disjuncts is false:

$$B = q \vee \neg p \vee \neg r \text{ is false} \iff q = 0,\ \neg p = 0,\ \neg r = 0.$$

Thus

$$q = 0, \quad p = 1, \quad r = 1.$$

Step 2: Check the antecedent under this assignment.
With $p = 1, q = 0, r = 1$,

$$A = q \vee \neg(p \wedge r) = 0 \vee \neg(1 \wedge 1) = 0 \vee \neg 1 = 0.$$

So whenever B is false, A is also false. Therefore there is *no* valuation with A true and B false. Hence the implication is never false, i.e. it is always true.

$$\boxed{\big(q \vee \neg(p \wedge r)\big) \rightarrow \big((q \vee \neg p) \vee \neg r\big) \text{ is a tautology.}}$$

Q3(a)

Prove the validity of the following argument.

If Mary runs for office, she will be elected. If Mary attends the meeting, she will run for office. Either Mary will attend the meeting or she will go to India. But Mary cannot go to India.

Thus Mary will be elected.

Solution

Encoding.

$$R: \text{ “Mary runs for office”}, \quad E: \text{ “Mary will be elected”},$$

$$M: \text{ “Mary attends the meeting”}, \quad I: \text{ “Mary goes to India”}.$$

Premises:

$$\begin{aligned}&(1)\quad R \to E,\\&(2)\quad M \to R,\\&(3)\quad M \vee I,\\&(4)\quad \neg I.\end{aligned}$$

Derivation (rule by rule).

1. $M \vee I$ (Premise 3)
2. $\neg I$ (Premise 4)
3. M (From 1–2 by *disjunctive syllogism* on $M \vee I$ and $\neg I$)
4. $M \to R$ (Premise 2)
5. R (From 3 and 4 by *modus ponens*)
6. $R \to E$ (Premise 1)
7. E (From 5 and 6 by *modus ponens*)

Thus E ("Mary will be elected") follows from the premises, so the argument is valid.

Q3(b)

Justify that the following premises are inconsistent.

(i) If Nirmala misses many classes through illness, then he fails high school.

(ii) If Nirmala fails high school, then he is uneducated.

(iii) If Nirmala reads a lot of books, then he is not uneducated.

(iv) Nirmala misses many classes through illness and reads a lot of books.

Solution

Encoding.

$$M: \text{ "Nirmala misses many classes through illness"},$$
$$F: \text{ "Nirmala fails high school"},$$
$$U: \text{ "Nirmala is uneducated"},$$
$$B: \text{ "Nirmala reads a lot of books"}.$$

Premises:

$$\begin{aligned}&(1)\quad M \to F,\\&(2)\quad F \to U,\\&(3)\quad B \to \neg U,\\&(4)\quad M \wedge B.\end{aligned}$$

Derivation of a contradiction.

1. $M \wedge B$ (Premise 4)
2. M (From 1 by simplification)

3. B (From 1 by simplification)
4. F (From 2 and 3 by modus ponens on $M \to F$)
5. U (From 4 and 5 by modus ponens on $F \to U$)
6. $\neg U$ (From 3 and 2 by modus ponens on $B \to \neg U$)
7. From 6 and 7 we obtain $U \wedge \neg U$, a contradiction.

Thus the set of premises forces both U and $\neg U$ to be true, which is impossible. Therefore the premises are **inconsistent**.

Q4(a)

Solve the following Boolean functions using K-map:

(i) $F(A, B, C, D) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$.

(ii) $F(A, B, C, D) = \Pi(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$.

Solution

We use 4-variable Karnaugh maps; variables are A, B for rows and C, D for columns in Gray code order $00, 01, 11, 10$.

(i) $F(A, B, C, D) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$
Filling the K-map and grouping:

- A group of four 1's in column $D = 1$ with $B = 1$ gives term BD.
- A group of four 1's in column $D = 0$ with $B = 0$ gives term $B'D'$.

Thus
$$F = BD + B'D'.$$
This is the XNOR of B and D.

$$\boxed{F(A, B, C, D) = BD + B'D'.}$$

(ii) $F(A, B, C, D) = \Pi(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$
Here F is given as a product of maxterms; i.e. $F = 0$ on those minterm indices. So $F = 1$ for minterms
$$\Sigma(3, 7, 10, 11, 15).$$
Plot these 1's on the K-map.

- A column of four 1's at $C = 1, D = 1$ gives term CD.
- A pair of 1's at minterms 10 and 11 (row $A = 1, B = 0$, columns $CD = 10, 11$) gives term $AB'C$.

Thus a minimal SOP is
$$\boxed{F(A, B, C, D) = CD + AB'C.}$$

Q4(b): K-map Simplification

Simplify the Boolean function:

$$F(A, B, C, D) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11).$$

Also draw the logic circuit of the simplified F.

Solution

First write the minterms in binary (with A as MSB):

m	$ABCD$			
0	0000	1 : 0001	2 : 0010	3 : 0011
4	0100	5 : 0101	6 : 0110	7 : 0111
8	1000	9 : 1001	11 : 1011	

Use a 4-variable K–map with rows $AB = 00, 01, 11, 10$ and columns $CD = 00, 01, 11, 10$.

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	1	1	1	0

Grouping:

- An octet covering all minterms with $A = 0$ (rows $AB = 00, 01$): this gives the term $\overline{A}$.
- A group of four cells $\{0, 1, 8, 9\}$ (rows $AB = 00, 10$, columns $CD = 00, 01$): here $B = 0$ and $C = 0$, so the term is $\overline{B}\,\overline{C}$.
- A group of four cells $\{1, 3, 9, 11\}$ (rows $AB = 00, 10$, columns $CD = 01, 11$): here $B = 0$ and $D = 1$, so the term is $\overline{B}D$.

So the minimized sum-of-products expression is

$$F = \overline{A} \ + \ \overline{B}\,\overline{C} \ + \ \overline{B}D.$$

Logic circuit for $F = \overline{A} + \overline{B}\,\overline{C} + \overline{B}D$:

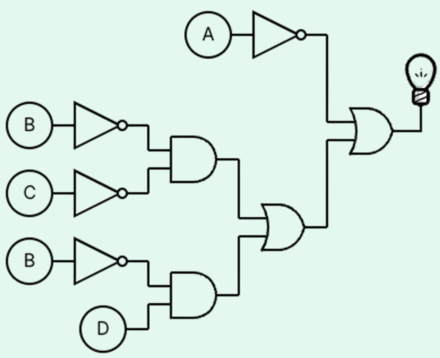

Figure 1: Final Circuit Diagram for F

Thus the correct simplified function is

$$\boxed{F(A, B, C, D) = \overline{A} + \overline{B}\,\overline{C} + \overline{B}D}$$

and the circuit above implements this expression.

Q5(a) Simplify the following Boolean expressions:

(i) $XY + (XZ)' + XY'Z(XY + Z)$
(ii) $C(B + C)(A + B + C)$

Solution

(i) Simplify $XY + (XZ)' + XY'Z(XY + Z)$

$$F = XY + (XZ)' + XY'Z(XY + Z)$$

Step 1: Expand the last term

$$XY'Z(XY + Z) = XY'Z \cdot XY + XY'Z \cdot Z = 0 + XY'Z = XY'Z$$

$$F = XY + (XZ)' + XY'Z$$

Step 2: Apply De Morgan's Theorem

$$(XZ)' = X' + Z'$$

$$F = XY + X' + Z' + XY'Z$$

Step 3: Combine terms

$$XY + XY'Z = X(Y + Y'Z)$$
$$Y + Y'Z = (Y + Y')(Y + Z) = Y + Z$$
$$XY + XY'Z = X(Y + Z) = XY + XZ$$

Thus:

$$F = X' + Z' + XY + XZ$$

Step 4: Apply Absorption

$$X' + XZ = X' + Z$$

So:

$$F = X' + Z + Z' + XY$$

$$Z + Z' = 1$$

$$F = X' + 1 + XY = 1$$

$$\boxed{F = 1}$$

(ii) Simplify $C(B + C)(A + B + C)$

$$F = C(B + C)(A + B + C)$$

Step 1: Simplify $C(B + C)$

$$C(B + C) = CB + C^2 = C(B + 1) = C$$

$$F = C(A + B + C)$$

Step 2: Simplify further

$$C(A+B+C) = AC + BC + C = C(A+B+1) = C$$

$$\boxed{F = C}$$

Q5(b):

Express $E(x, y, z) = x(y'z)'$ in its complete sum-of-products form.

Solution

Given:

$$E(x, y, z) = x(y'z)'$$

Step 1: Simplify the expression

$$(y'z)' = (y')' + z' = y + z' \quad \text{(De Morgan's law)}$$

$$E = x(y + z') = xy + xz'$$

Step 2: Convert to canonical (complete) SOP

For xy:

$$xy = xy(z + z') = xyz + xyz'$$

For xz':

$$xz' = xz'(y + y') = xyz' + xy'z'$$

Therefore:

$$E = xyz + xyz' + xyz' + xy'z' = xyz + xyz' + xy'z'$$

Step 3: Write as sum of minterms

$$xy'z' \equiv m_4, \quad xyz' \equiv m_6, \quad xyz \equiv m_7$$

$$\boxed{E(x, y, z) = xyz + xyz' + xy'z' = \sum m(4, 6, 7)}$$